

555 Timer Design Using MATLAB



NISHANT MITTAL

555 timer

Presented here is a MATLAB program to design an astable multivibrator of desired frequency and duty cycle using a 555 timer. The aesthetic graphical user interface of the program helps you quickly calculate the values of timing components for your design requirements.

Fig. 1 shows the graphical interface where the desired frequency and duty cycle can be directly entered to calculate the value of timing components by the program.

Let us first discuss the basic functionality of a 555 timer to better understand how the MATLAB program works.

The 8-pin 555 timer is one of the most popular ICs in the electronics world. With just a few external components it can be used to build amazing timer, pulse generator and oscillator applications. A popular version is NE555, which is suitable in most cases where a '555 timer' is specified.

Fig. 2 shows the internal diagram of a 555 timer to help you understand its pins' functionalities:

Pin 1: Ground terminal. All the voltages are measured with respect to this terminal.

Pin 2: Trigger terminal. This pin is an inverting input to comparator-2. It

pulse is applied. When this pin is not to be used for reset purpose, it should be connected to +Vcc to avoid any possibility of false triggering.

Pin 5: Control voltage terminal. The function of this terminal is to control the threshold and trigger levels.

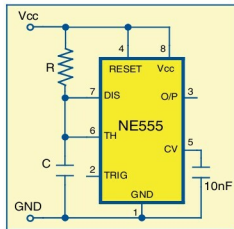


Fig. 3: 555 timer in monostable mode

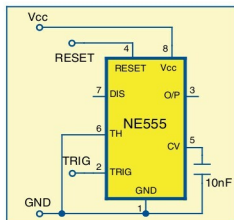


Fig. 4: 555 timer in bistable mode

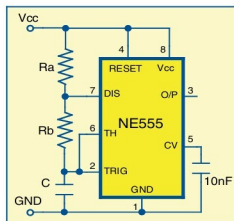


Fig. 5: Astable mode of 555 timer



Fig. 1: GUI 555 timer designer

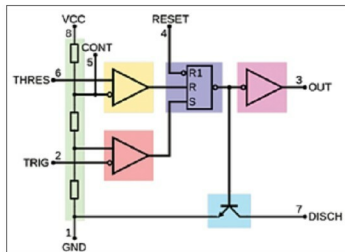


Fig. 2: Internal diagram of a 555 timer

is responsible for transition of the output of flip-flop from set to reset. The output of the timer depends on the amplitude of the external trigger pulse applied to this pin.

Pin 3: Output terminal. The output of the timer is available at this pin.

Pin 4: Reset terminal. To disable or reset the timer, a negative